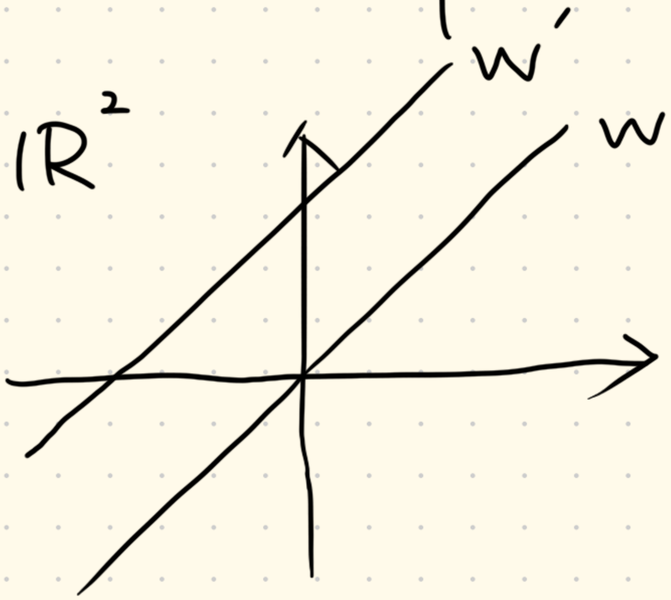
Linear spaces and subspaces.



$M_{m\times n}(\mathbb{R})$

upper triangular matrix.

lower triangular matrix.

symmetric matrix

skew - symmetric matrix

$$\mathbb{R}[x] = \{ a_0 + a_1 x + \cdots + a_n x^n : n \geq 0,\ a_i \in \mathbb{R} \}.$$

$$P_n = \{ p(x) \in \mathbb{R}[x] : \deg p \leq n \}.$$

$$S = \{ p(x) \in \mathbb{R}[x] : \deg p = n \} \text{ is not a subspace.}$$

$$\mathcal{F}(\mathbb{R}, \mathbb{R}), \quad (f+g)(x) = f(x) + g(x), \quad (\lambda f)(x) = \lambda f(x).$$

$C(\mathbb{R})$ continuous functions.

$C^k(\mathbb{R})$ k-times differential functions.

$C^\infty(\mathbb{R})$ smooth functions.

odd functions. even functions.

periodic functions $f(x+T) = f(x)$, $T > 0$.

$C([0,1])$

$W := \{ f \in C([0,1]) : \int_0^1 f(x)\,dx = 0 \}$.

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + C = 0. \qquad \text{solutions}$$

$y'' + y = 0$ $\quad \{ a\cos x + b\sin x : a, b \in \mathbb{R} \}$

$y'' + y = 1$ $\quad \{ a\cos x + b\sin x + 1 : a, b \in \mathbb{R} \}$ not.

$$\mathbb{R}^{\mathbb{N}} := \{(a_1, a_2, a_3, \dots) : a_i \in \mathbb{R}\}.$$

bounded sequences. $\sup_n |a_n| < \infty$.

convergent sequences. $\lim_{n \to +\infty} a_n = a$.

$T : V \to W$ linear. $\ker T$. $\operatorname{im} T$ subspaces

U, W subspaces. $U + W$. $U \cap W$ subspaces.